Exp 5 · Magnitude comparator (1-bit and 7485) and odd-parity generator/checker (gates and 74180)

Manual pp. 13–18 · ICs 7404, 7408, 7486, 7485, 74180

MANUAL EXP 5, AIM

Design and study 1-bit and 4-bit (7485) magnitude comparators, and a 3-bit odd-parity generator/checker, then the 74180. Viva form: "A lone 7485 lights both A>B and A<B when the inputs are equal. What did you forget?"

GUESS FIRST

The 7485 has three *cascade inputs*. What must they be set to when only one chip is used?

1-bit comparator

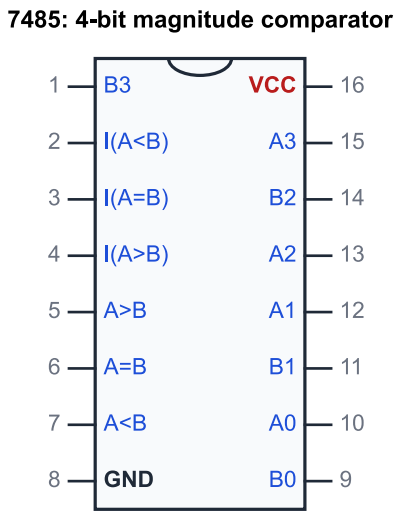
$$(A > B) = AB' \qquad (A < B) = A'B \qquad (A = B) = (A \oplus B)'$$

Wiring:

- 7404: A → 1 → 2 (A'); B → 3 → 4 (B').
- 7408: A → 1, 7404-4 → 2 → **3 = (A > B)**; 7404-2 → 4, B → 5 → **6 = (A < B)**.
- 7486: A → 1, B → 2 → 3; 7404: 7486-3 → 5 → **6 = (A = B)**.

Only one output is HIGH at a time.

4-bit comparator, 7485



7485. The cascade inputs I(A>B), I(A=B), I(A

Signal	A3 A2 A1 A0	B3 B2 B1 B0	I(A>B)	I(A=B)	I(A<B)	A>B, A=B, A<B out
pin	15, 13, 12, 10	1, 14, 11, 9	4 → **GND**	3 → **+5 V**	2 → **GND**	5, 6, 7

The chip compares from the MSB down. The first unequal bit decides the result. Only when all four bit pairs are equal does it pass the **cascade inputs** through. So a single chip needs I(A=B) = 1 (and the others 0) for "equal" to come out right.

8-bit (exercise):

- The **low** nibble's chip has cascade inputs 0, 1, 0.
- Its three outputs (pins 5, 6, 7) go into the **high** nibble chip's cascade inputs (pins 4, 3, 2).
- The final answer comes from the high chip.

Odd parity generator and checker

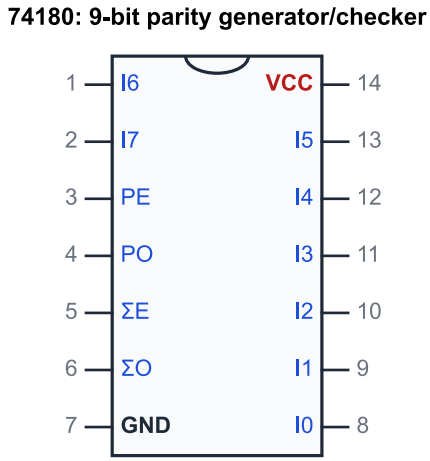
Odd parity makes the total number of 1s (data + parity) **odd**.

- **Generator:** P = (X ⊕ Y ⊕ Z)'. Two XORs (7486) and an inverter (7404). The manual's Fig 5.4 feeds Y, Z into gate 1 and X with gate 1's output into gate 2.
- **Checker:** C = (X ⊕ Y ⊕ Z ⊕ P)'. Three XORs and an inverter. **C = 1 flags an error** (an even count arrived).

Manual Fig 5.5 puts inverters after the first two XORs. Those two inversions cancel inside the last XOR, so the result is the same.

XYZ	000	001	010	011	100	101	110	111
P (odd)	1	0	0	1	0	1	1	0

74180: 9-bit parity generator/checker



74180. Data I0–I7; enables PE (pin 3) and PO (pin 4);
outputs ΣE (pin 5) and ΣO (pin 6).

Count of 1s in I0–I7	PE	PO	ΣE	ΣO
even	1	0	1	0
odd	1	0	0	1
even	0	1	0	1
odd	0	1	1	0
any	1	1	0	0
any	0	0	1	1

The manual's observation table, filled (checked):

I7…I0	PE PO	ΣE ΣO
1111 0000	1 0	**1 0**
1111 0001	1 0	**0 1**
1010 1100	0 1	**0 1**
1010 1101	0 1	**1 0**
0000 0000	1 0	**1 0**
1111 1111	0 1	**0 1**

WRITE IN YOUR RECORD

Result: the 1-bit comparator and the 7485 4-bit comparator gave exactly one HIGH output per input pair. The 3-bit odd-parity generator/checker was verified (checker output 1 = error). The 74180 outputs matched the observation table.

Theory for the record: a magnitude comparator decides A > B, A = B or A < B. For one bit: AB', A'B and (A ⊕ B)'. The 7485 compares two 4-bit words from the MSB down. Its cascade inputs let chips be chained for longer words, and must be set to (0, 1, 0) for a single chip. A parity bit makes the number of 1s even or odd so that single-bit errors can be detected. Odd-parity generator: P = (X ⊕ Y ⊕ Z)'. The checker XORs all received bits. The 74180 generates/checks parity on 8 bits, with PE/PO selecting even/odd.

CHECK 1

(a) A = 1010, B = 1010 on a lone 7485 with I(A=B) left floating. What do you expect, and why is "floating" risky here? (b) Odd parity, data 101: give P, and the checker output if the received word is 1011. (c) How do you make a 9-bit odd-parity checker from one 74180 and an inverter?

Answers

Opening question. The cascade inputs: I(A=B) (pin 3) must be **HIGH**, and I(A>B), I(A<B) (pins 4, 2) **LOW**.

Check 1. (a) A floating TTL input usually acts HIGH, so it may happen to work. That relies on undefined behaviour, so tie pin 3 to +5 V. The datasheet's function table shows what wrong cascade settings do when A = B: all three at 0 gives A>B = A**B) = I(A<B) = 1 with I(A=B) = 0 gives all three outputs 0. (b) P = (1 ⊕ 0 ⊕ 1)' = 1. The word 1011 has three 1s (odd) = valid, so the checker output is 0 (no error). (c) Feed 8 bits to I0–I7. Feed the 9th bit to PO, and its inverse (via the inverter) to PE. Then ΣO/ΣE give the odd-parity check of all 9 bits.**